

Demostración alternativa más general:

línea l y vector $\rightarrow$ combinación lineal más general de 2 vectores que de un vector es $u \cdot G \cdot v = G_{ij} u_i v_j$

$$\underline{A} \times (\underline{B} \times \underline{C}) = a(\underline{A}, \underline{C}) \underline{B} + b(\underline{A}, \underline{B}) \underline{C}$$

nueve componentes independientes en el caso + general

$$a(\underline{A}, \underline{C}) = \underline{A} \cdot \underline{\alpha} \cdot \underline{C} = \alpha_{ij} A_i C_j$$

$$b(\underline{A}, \underline{B}) = \underline{A} \cdot \underline{\beta} \cdot \underline{B} = \beta_{ij} A_i B_j$$

Son 18 incógnitas que hay que determinar

(para ello se debe plantear 18 ecuaciones obteniendo los valores $i \neq k$; las combinaciones obtenidas sería suficiente)

$$\underline{A} \times (\underline{B} \times \underline{C}) = (\underline{A} \underline{\alpha} \underline{C}) \cdot \underline{B} + (\underline{A} \underline{\beta} \underline{B}) \underline{C}$$

$$(*) \quad \underline{e}_i \cdot \underline{G} \cdot \underline{e}_j = (\underline{e}_i)_p G_{pq} (\underline{e}_j)_q = \delta_{ip} G_{pq} \delta_{jq} = G_{ij}$$

$$\underline{e}_i = \underline{e}_1 = \underline{i} \quad \underline{e}_j = \underline{j} \quad (\text{direct}) \quad \underline{e}_j = \underline{j} \quad (\text{direct})$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \\ G_{31} & G_{32} & G_{33} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} G_{12} \\ G_{22} \\ G_{32} \end{bmatrix} = G_{12}$$

• Si $\underline{B} = \underline{i}, \underline{C} = \underline{j} \wedge \underline{A} = \underline{i}$:

$$\underline{A} \times (\underline{B} \times \underline{C}) = \underline{i} \times (\underline{i} \times \underline{j}) = (\underline{i} \underline{\alpha} \underline{j}) \underline{i} + (\underline{i} \underline{\beta} \underline{i}) \underline{j}$$

$$\underline{v} = \alpha_{12} \underline{i} + (1 + \beta_{11}) \underline{j} = \underline{0} \Rightarrow \alpha_{12} = 0$$

$$1 + \beta_{11} = 0 \Rightarrow \beta_{11} = -1$$

• Si $\underline{B} = \underline{i}, \underline{C} = \underline{j} \wedge \underline{A} = \underline{j}$:

$$\underline{A} \times (\underline{B} \times \underline{C}) = \underline{j} \times (\underline{i} \times \underline{j}) = \underline{i} = \alpha_{22} \underline{i} + \beta_{21} \underline{j} \Rightarrow (\alpha_{22} - 1) \underline{i} + \beta_{21} \underline{j} = \underline{0}$$

$$\Downarrow$$

$$\alpha_{22} = 1$$

$$\beta_{21} = 0$$

• Si $\underline{B} = \underline{i}$, $\underline{C} = \underline{d}$ y $\underline{A} = \underline{k}$

$$\underline{k} \times (\underbrace{\underline{i} \times \underline{d}}_{\underline{k}}) = \underline{0} = \underline{\alpha}_{32} \cdot \underline{i} + \underline{\beta}_{31} \underline{d} \Rightarrow \begin{cases} \alpha_{32} = 0 \\ \beta_{31} = 0 \end{cases}$$

Siguiente con esta operacio se obtiene que

$$\underline{\alpha} = \underline{I}$$

$$\underline{\beta} = -\underline{I}$$

$$\underline{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(\underline{I})_{ij} = \delta_{ij}$$

$$(*)*) \quad (\underline{u} \cdot \underline{I} \cdot \underline{v} = u_i \delta_{ij} v_j = u_i v_i = \underline{u} \cdot \underline{v})$$

$$\underline{A} \cdot \underline{\alpha} \cdot \underline{C} = \underline{A} \cdot \underline{C}$$

$$\underline{A} \cdot \underline{\beta} \cdot \underline{B} = -\underline{A} \cdot \underline{B}$$

$$\left. \begin{array}{l} \underline{A} \cdot \underline{\alpha} \cdot \underline{C} = \underline{A} \cdot \underline{C} \\ \underline{A} \cdot \underline{\beta} \cdot \underline{B} = -\underline{A} \cdot \underline{B} \end{array} \right\} \underline{A} \times (\underline{B} \times \underline{C}) = (\underline{A} \cdot \underline{C}) \underline{B} - (\underline{A} \cdot \underline{B}) \underline{C}$$